CENG 252: Chemical Reaction Engineering

Practice Midterm 2 (Lectures 6–10)

Time: 1 hour 20 minutes • Total: 100 points

Instructions

- Closed book, closed notes, no cheat sheet. A formula sheet is provided on the last page. Only a simple scientific calculator.
- **Show all work.** For mechanisms: write every elementary step's rate law, state which species each k refers to, apply the PSSH or the equilibrium of the fast steps, and write the conservation (enzyme or site) balance *before* simplifying. Partial credit is given for a correct setup.
- State every assumption. Box your final answers and include units.
- Budget your time: about 0.8 minutes per point (16 minutes for a 20-point problem).

Problem	Topic	Points	Score
1	Short answers and concepts (all lectures, incl. Lec. 10)	25	
2	Active intermediates and the PSSH: ozone decomposition	15	
3	Enzyme kinetics, inhibition and a batch bioreactor	20	
4	Heterogeneous catalysis: mechanism, data, reactor and deactivation	40	
	Total	**100**	

Constants: $R = 8.314$ J/(mol·K) $= 1.987$ cal/(mol·K) $= 0.08206$ dm^3·atm/(mol·K).

Problem 1: Short answers and concepts

[25 pts]

(a) [3] The termination step of a free-radical chain is $2\,CH_3^{\bullet} \xrightarrow{k_5} C_2H_6$, with k_5 defined with respect to C_2H_6, so $r_{5,C_2H_6} = k_5 C_{CH_3\bullet}^2$. Write the term that this step contributes to the PSSH balance for $CH_3\bullet$. Explain the factor.

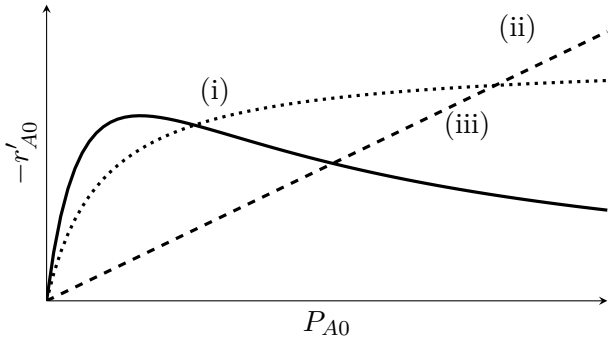
(b) [3] A catalytic hydrogenation $A + H_2 \longrightarrow B$ has the measured rate law

$$-r'_A = \frac{k\,P_A\,P_{H_2}^{1/2}}{\left(1 + K_A P_A + \sqrt{K_H P_{H_2}}\right)^2}.$$

Give three conclusions about the mechanism that you can read directly from this expression.

(c) [3] Two inhibitors each lower the rate of an enzyme reaction. With inhibitor X, the Lineweaver–Burk line keeps the same intercept but becomes steeper. With inhibitor Y, the line shifts up parallel to the original. Name each type. For which one can the original V_{max} still be reached by adding a large excess of substrate? Why?

(d) [3] The initial rate of A $\longrightarrow$ B over a solid catalyst was measured against the feed pressure of pure A for three different catalysts. Match each curve (i), (ii), (iii) to its rate-limiting step (adsorption of A, single-site surface reaction, or dual-site surface reaction), and give one sentence of justification for each.



(e) [3] The activity of a reforming catalyst was measured as $a = 0.80$ after 10 h and $a = 0.50$ after 40 h. Show whether first-order or second-order decay fits these data. Name the deactivation mechanism that usually gives this order, and find the time at which $a = 0.25$.

(f) [3] Explain what *separable* deactivation kinetics means. Write the rate law of a decaying catalyst in separable form, define the activity $a(t)$, and state why the separable assumption is useful. Why do commercial reactors slowly raise their temperature over a catalyst's life?

(g) [3] For each situation, choose the most suitable reactor (packed bed, moving bed, or straight-through transport reactor) and give a one-line reason:

(i) an Fe catalyst for NH_3 synthesis that stays active for several years;

(ii) a cracking catalyst that is covered with coke within a few seconds;

(iii) a catalyst that loses most of its activity over several hours and must be regenerated continuously without shutting the plant down.

(h) [4] For each case, name the deactivation mechanism (sintering, poisoning, or fouling/coking), say whether it is reversible, and suggest one way to prevent or reduce it:

(i) a 13.5% Ni/SiO_2 catalyst held at 750°C loses about 70% of its Ni surface area in 50 h;

(ii) a Ni steam-reforming catalyst is fed natural gas containing a few ppm of H_2S;

(iii) carbonaceous deposits cover the active sites of a zeolite cracking catalyst and plug its pores.

Also explain in one sentence how O_2 can be a *reactant* for ethylene oxidation on Ag but a *poison* for ethylene hydrogenation on Ni.

Problem 2: Active intermediates: ozone decomposition

[15 pts]

The gas-phase decomposition $2\,O_3 \longrightarrow 3\,O_2$ is believed to go through atomic oxygen, O, as an active intermediate:

$$(1)\ O_3 \xrightarrow{k_1} O_2 + O \qquad (2)\ O_2 + O \xrightarrow{k_2} O_3 \qquad (3)\ O + O_3 \xrightarrow{k_3} 2\,O_2$$

All rate constants are defined with respect to the species O. Every step is elementary.

(a) [6] Use the PSSH to derive the rate of disappearance of ozone, $-r_{O_3}$, in terms of measurable concentrations only.

(b) [4] Experiments carried out in a large excess of O_2 give $-r_{O_3} = k_{obs}\dfrac{C_{O_3}^2}{C_{O_2}}$. State the condition under which your result reduces to this form, and express k_{obs} in terms of k_1, k_2, k_3. What rate law do you expect at the very start of a run that begins with pure ozone?

(c) [3] The activation energies of the elementary steps are $E_1 = 100$ kJ/mol, $E_2 = 2$ kJ/mol and $E_3 = 20$ kJ/mol. Find the apparent activation energy of k_{obs}. By what factor does k_{obs} increase from 300 K to 310 K?

(d) [2] At a fixed, moderate O_2 concentration, sketch $-r_{O_3}$ versus C_{O_3} over a wide range of C_{O_3}, and label the apparent reaction order in each region. Which elementary step limits the rate at high C_{O_3}?

Problem 3: Enzyme kinetics, inhibition and a batch bioreactor

[20 pts]

An enzyme E converts substrate S to product P by the Michaelis–Menten mechanism $\mathrm{E} + \mathrm{S} \rightleftharpoons \mathrm{E}\cdot\mathrm{S} \longrightarrow \mathrm{E} + \mathrm{P}$. The initial rates below were measured at the same total enzyme concentration, first with no inhibitor and then with an inhibitor I at $C_I = 4.0$ mmol/L.

C_S (mmol/L)	1.0	2.0	5.0	10.0	20.0
$-r_S$, no inhibitor (mmol/L/min)	0.333	0.571	1.000	1.333	1.600
$-r_S$, with $C_I = 4.0$ mmol/L (mmol/L/min)	0.250	0.364	0.500	0.571	0.615

(a) [6] In *uncompetitive* inhibition, I binds only to the enzyme–substrate complex: $\mathrm{E}\cdot\mathrm{S} + \mathrm{I} \rightleftharpoons \mathrm{I}\cdot\mathrm{E}\cdot\mathrm{S}$ (inactive), with $K_I = C_{E\cdot S} C_I / C_{I\cdot E\cdot S}$. Starting from the mechanism and the total enzyme balance, derive $-r_S$ for uncompetitive inhibition and write it in Lineweaver–Burk (reciprocal) form. State what happens to the slope and the intercept compared with no inhibition.

(b) **[5]** Using a Lineweaver–Burk analysis of the uninhibited data, find $V_{\max}$ and K_M (with units).

(c) **[4]** Analyse the inhibited data the same way. Which type of inhibition is this? Find K_I.

(d) [5] The same enzyme loading is used in a well-mixed, constant-volume batch reactor with $C_{S0} = 20$ mmol/L. Starting from the mole balance, derive an expression for the time needed to reach conversion X, and compute the time for $X = 0.90$ (i) with no inhibitor and (ii) with $C_I = 4.0$ mmol/L held constant. Explain in one sentence why the inhibitor has almost no effect late in the batch, when C_S is small.

Problem 4: Heterogeneous catalysis: from mechanism to reactor

[40 pts]

The gas-phase dehydration of an alcohol A to an olefin B and water W, $\mathrm{A} \longrightarrow \mathrm{B} + \mathrm{W}$, is carried out over a solid acid catalyst at 500 K. The proposed mechanism is

(1) $\mathrm{A} + \mathrm{S} \rightleftharpoons \mathrm{A}\cdot\mathrm{S}$ (2) $\mathrm{A}\cdot\mathrm{S} + \mathrm{S} \longrightarrow \mathrm{B}\cdot\mathrm{S} + \mathrm{W}\cdot\mathrm{S}$ (3) $\mathrm{B}\cdot\mathrm{S} \rightleftharpoons \mathrm{B} + \mathrm{S}$ (4) $\mathrm{W}\cdot\mathrm{S} \rightleftharpoons \mathrm{W} + \mathrm{S}$

Step (2) is irreversible. Use adsorption equilibrium constants K_A, K_B, K_W, and C_t for the total site concentration.

(a) [8] Derive the rate law $-r'_A$ assuming the surface reaction (2) is rate-limiting. Show the site balance.

(b) [6] Derive the rate law assuming instead that the adsorption of A (step 1) is rate-limiting. (Hint: what does "fast and irreversible" imply for C_{AS}?) Write the initial rate for each of the two cases, (a) and (b).

(c) [9] Initial rates measured with pure A in the feed (no B or W present) are:

P_{A0} (atm)	0.5	1.0	2.0	4.0	8.0
$-r'_{A0}$ (mol/kg cat/s)	0.160	0.222	0.250	0.222	0.160

Which of your rate laws, (a) or (b), is consistent with these data? Explain. Using a suitable linearisation, determine the lumped rate constant k (with units) and K_A.

(d) [9] Separate experiments show that B and W are only weakly adsorbed ($K_B P_B$, $K_W P_W \ll 1$). The reaction is run in a fluidised-bed reactor that behaves as a CSTR, isothermal at 500 K with no pressure drop. Pure A is fed at $F_{A0} = 10$ mol/s.

(i) Find the catalyst mass needed for $X = 0.50$ when the inlet pressure is $P_0 = 2$ atm.

(ii) An engineer suggests that a higher pressure always means a faster rate. At the same F_{A0} and $X = 0.50$, what inlet pressure *minimises* the catalyst mass, and what is that minimum mass? Explain physically why raising the pressure further does not help.

(e) [8] The reactor is now operated at the optimum from (d)(ii). The catalyst sinters, following second-order decay with $k_d = 0.0125\ \text{h}^{-1}$. The fluidised bed is well mixed, so all of the catalyst has the same age.

(i) Find the activity after 80 h. To what value must F_{A0} be reduced (same P_0) to keep $X = 0.50$?

(ii) Instead of reducing the feed, the operators keep $F_{A0} = 10$ mol/s and raise the temperature. The rate constant k has $E = 80$ kJ/mol. Assume that K_A and k_d do not change with T, and that the inlet partial pressure is held fixed. What temperature is needed after 80 h?

(iii) Give one reason why the strategy in (ii) shortens the catalyst life in practice.

Formula sheet (provided)

Relative rates $aA + bB \to cC + dD$: $\dfrac{r_A}{-a} = \dfrac{r_B}{-b} = \dfrac{r_C}{c} = \dfrac{r_D}{d}$

PSSH $r_{A^*} = \sum_i r_{iA^*} = 0$

Arrhenius $k(T) = k(T_0)\exp\left[\dfrac{E}{R}\left(\dfrac{1}{T_0} - \dfrac{1}{T}\right)\right]$

Design equations batch (constant V): $\dfrac{dC_S}{dt} = r_S$ CSTR: $W = \dfrac{F_{A0}X}{-r'_A}$ PBR: $F_{A0}\dfrac{dX}{dW} = -r'_A$

Gas stoichiometry $P_A = P_{A0}\dfrac{1-X}{1+\varepsilon X}\,y\,\dfrac{T_0}{T}, \qquad \varepsilon = y_{A0}\delta$

Michaelis–Menten $-r_S = \dfrac{V_{\max}C_S}{K_M + C_S}, \qquad V_{\max} = k_{\text{cat}}C_{E,t}$

Catalytic step rates

adsorption: $r_{AD} = k_A\left(P_A C_v - \dfrac{C_{\text{AS}}}{K_A}\right)$

surface reaction: $r_S = k_S\left(C_{\text{AS}} - \dfrac{C_{\text{BS}}}{K_S}\right)$

desorption: $r_D = k_D\left(C_{\text{BS}} - K_B P_B C_v\right)$

site balance: $C_t = C_v + \sum_i C_{i\cdot S}$

Catalyst activity $-r'_A = a(t)\,(-r'_A)_{\text{fresh}}, \qquad -\dfrac{da}{dt} = k_d a^n$

Integrals $\displaystyle\int \frac{dx}{x} = \ln x, \qquad \int \frac{dx}{x^2} = -\frac{1}{x}, \qquad \int_0^X \frac{dX}{1-X} = \ln\frac{1}{1-X}$